## Table 1. Data sources and their roles

| Dataset / endpoint | Cases | Controls | Role |
|---|---|---|---|
| eQTLGen blood cis-eQTL | — | — | Exposure; up to 31,684 participants (median 28,092 per variant); predominantly European |
| BBJ GD (GCST90018627) | 2,809 | 172,656 | Discovery; East Asian |
| FinnGen GD (E4_GRAVES_STRICT) | 3,962 | 496,386 | Post hoc same-disease comparison; Finnish |
| UKB hyperthyroidism (GCST90038636) | 3,731 | 480,867 | Broader-phenotype comparison; self-reported hyperthyroidism or thyrotoxicosis; ancestry not established as exclusively European |
| FinnGen GO (GRAVES_OPHT) | 858 | 499,490 | Original ophthalmopathy (TED-enriched) comparison; Finnish |
| FinnGen GO strict (E4_GRAVES_OPHT_STRICT) | 753 | 499,595 | Post hoc ophthalmopathy definition sensitivity; Finnish |
| GTEx v8 tissue eQTL | — | — | Post hoc tissue extension (thyroid, cultured fibroblasts, subcutaneous and visceral adipose, whole blood; n = 469–670; Table S4) |
| 1000 Genomes Phase 3 European reference (n = 503) | — | — | LD clumping of instruments and reference allele frequencies |

The three FinnGen R12 endpoints share participants and are not independent replications. FinnGen ophthalmopathy controls are all other participants, including people with GD but no recorded eye disease, so GO estimates do not estimate TED risk among patients with GD. UKB hyperthyroidism is broader than GD. ‘Strict’ is the FinnGen endpoint label, not proof of perfect specificity. The descriptive orbital RNA-sequencing series (four TED patients, one control) is separate from these datasets. eQTL, expression quantitative trait locus; GTEx, Genotype-Tissue Expression; LD, linkage disequilibrium; TED, thyroid eye disease; BBJ, Biobank Japan; FinnGen GD, E4_GRAVES_STRICT; GD, Graves disease; GO, Graves ophthalmopathy; UKB, UK Biobank.